

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech CST 3rd Semester Final Examinations, November-December 2023

Data Structures CS-2103

Full Marks: 50

Time: 3 hours

Attempt all questions

1. Answer **all** of the following questions

a) Write down the following -

- i) The postfix form of the infix expression: $D * (A - B) + F / (E * A + G) + K$
- ii) The infix form of the postfix expression: $A B - C D + *$

b) Suppose a hash table T has 12 slots, $T[0], T[1], \dots, T[11]$ where all the slots are empty as shown below-

-	-	-	-	-	-	-	-	-	-	-	-
0	1	2	3	4	5	6	7	8	9	10	11

Also suppose 9 keys A, B, C, D, E, F, G, H, and I with the following hash addresses appeared

Keys:	A	B	C	D	E	F	G	H	I
$h(k)$:	3	6	11	11	3	10	4	2	3

Find the following and give brief explanation –

- i) Insert the keys into the hash table T as per the order shown above. If a collision occurs, follow the open addressing scheme to deal with it and draw the final hash table.
- ii) Average number of probes(comparison) for a successful search (S)
- iii) Average number of probes(comparison) for an unsuccessful search (U)
- iv) The load factor (λ) of the hash table T.

c) Do the following for a Min-heap (ascending heap) with a brief explanation –

- i) Create a min-heap by inserting the following keys [26, 19, 57, 61, 11, 49, 15, 8] in that order. Show the resulting min-heap after inserting each element.
- ii) Show the heap on performing delete operation (removing the root) **twice**.

(4+7+7 = 18)

2. Answer **any 2** of the following questions using 'C' programming language

- a) Suppose L_1 and L_2 are two sorted singly linked lists of integers. Write down a function to merge them together to create another singly sorted linked list L_3 .
- b) Assume there is a binary tree stores integer elements and is implemented using the linked-nodes approach. Write down the binary tree node structure. Also write down a function to print all the internal nodes (non-leaf nodes) of the binary tree.
- c) You are given two linked lists L_1 and L_2 , sorted in increasing order. Write down a function to create and return a new linked list L_3 representing the intersection (i.e. nodes with common values) of the two lists.

(2x6 = 12)

3. Answer **any 5** of the following questions

- a) Calculate the address of a random element present at $(i, j)^{th}$ location in a 2-D array of size $m \times n$, given the base address as α .
- b) Prove that a perfect binary tree of height h has $2^{(h+1)} - 1$ nodes.
- c) Draw the Binary Search Tree (BST) on inserting following elements [5, 2, 4, 1, 9, 3, 6, and 7] in that order into an initially empty BST.
Explain the steps to delete 6 and then 2 from the created BST.
- d) Write down a C function to check whether a given binary tree is full (strict) or not
- e) Explain the technique to perform quick sort.
What is the role of pivot element in the context of efficiency of the sorting?
- f) Mention properties of good hash function in the context of hashing.
How do you map a hash value to respective hash buckets?
- g) Write a short note on "AVL Tree".

(5x4 = 20)